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a.

Ans. to the q. no $\rightarrow 1$

The bitwise AND operation is used to determine whether a line segment is completely outside the clipping window and or partially inside. If the two bit position has 1 in their opcode then it means they are in same side and can not enter the window.

b.

$$P(t) = P_0 + t(P_1 - P_0)$$

$$= (1, 3) + 0.3((7-1), (10-3))$$

$$= (1, 3) + (0.3 \times 6, 0.3 \times 7)$$

$$= (2.8, 5.1)$$

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$$\therefore t_F = 0.3529 \quad t_L = 1$$

$$P(0.35) = -50 + 0.35 \times 90, -70 + 0.35 \times 170$$

$$= (-18.25, -10.25)$$

$$P(1) = (40, 100)$$

B.

cohen sutherland :

$$p_0 \rightarrow 0100$$

$$p_1 \rightarrow 0000$$

$$m = \frac{40 \cdot 100 + 70}{40 + 100}$$

$$= 1.889$$

bottom line clipping

$$y = y_{\min} = -10$$

$$m = \frac{y - y_1}{x - x_1}$$

$$\Rightarrow 1.89 = \frac{y_{\min} + 70}{x + 50}$$

$$\Rightarrow 1.80 = \frac{-10 + 70}{2 + 50}$$

$$\Rightarrow (1.80x + 24.5 = 60) \text{ origin of element (i)}$$

$$\Rightarrow x = -18.25 \text{ (e, e) 2 failed (ii)}$$

$$y = -10 \text{ (e, 25) T and element (iii)}$$

so clipped point $(-18.25, -10), (40, 100)$

Ans. to the q. no 2

Reflection about coordinate axes are simple.
 x axis: $(x, y) \rightarrow (x, -y)$

y axis: $(x, y) \rightarrow (-x, y)$

But reflection about arbitrary line is more complex. There is no simple pattern for it.
 So we need to translate that line to pass through the origin and then rotate it to align with an axis then apply reflection then rotate back and translating back.

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b.

Scaling :

i) translate to origin $T(-75, -25)$

ii) scaling $S(2, 2)$

iii) translate back $T(75, 25)$

$$M_{SC} = T(75, 25) \times S(2, 2) \times T(-75, -25)$$

$$= \begin{bmatrix} 1 & 0 & 75 \\ 0 & 1 & 25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -75 \\ 0 & 1 & -25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -75 \\ 0 & 2 & -25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 75 \\ 0 & 1 & 25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -150 \\ 0 & 2 & -50 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 70 \\ 50 \\ 1 \end{bmatrix}$$

$$\therefore P \begin{bmatrix} x \\ y \\ z \end{bmatrix} = M_S \cdot P_{T^{-1}} \cdot P_{T^{-1}}^{-T} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -75 \\ 0 & 2 & -25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 70 \\ 50 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 65 \\ 75 \\ 1 \end{bmatrix}$$

Rotation:

- i) Translate to origin
- ii) Rotate
- iii) Translate back

$$T_p = \begin{bmatrix} 1 & 0 & -50 \\ 0 & 1 & -50 \\ 0 & 0 & 1 \end{bmatrix}$$

x + 50
y + 50
z

$$R = \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_p^{-1} = \begin{bmatrix} 1 & 0 & 50 \\ 0 & 1 & 50 \\ 0 & 0 & 1 \end{bmatrix}$$

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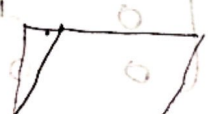
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$$P_2' \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = T_p^{-1} \cdot R \cdot T_p \cdot P = \begin{bmatrix} 1 & 0 & -50 \\ 0 & 1 & -50 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -50 \\ 0 & 1 & -50 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 42.93 \\ 78.28 \\ 1 \end{bmatrix}$$

Translation to (i)
Rotate (ii)
Translation back (ii)

Shear :

$$\begin{aligned} x &\rightarrow x + sh_x y \\ y &\rightarrow y + sh_y x \end{aligned} \quad \begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & sh_y \\ 0 & 0 & 1 \end{bmatrix} = T_s$$


$$P' \begin{bmatrix} 100 \\ 150 \\ 1 \end{bmatrix} = T_h^{-1} \cdot H \cdot T_h \cdot P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & sh_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 42.93 \\ 78.28 \\ 1 \end{bmatrix}$$

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$$= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & shx & 0 \\ shy & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 32.93 \\ 68.28 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + shy & shx & 10 \\ shy & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 32.93 \\ 68.28 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 32.93 + 68.28 shx + 10 \\ 32.93 shy + 68.28 + 10 \\ 1 \end{bmatrix}$$

$$\therefore 42.93 + 68.28 shx = 100$$

$$\therefore shx = 0.836$$

$$\therefore 32.93 shy + 78.28 = 150$$

$$\therefore shy = 2.178$$